ARJUNA JEE 2027

Lecture -07 | Mathematics

PERMUTATIONS AND COMBINATIONS

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Topics *to be covered*

A Rank of a Word in Dictionary

B Selection out of things which are not all different

C Question Practice

D



LAST CLASS RECAP

Consider four different books of Mathematics, three different books of Physics and two different books of Chemistry. Number of ways of arranging all nine books in a row such that:

(i) All the books of same subject are together is $3!_0 \cdot 4!_0 \cdot 3!_0 \cdot 2!_0$.

(ii) No two books of Mathematics are together is $\left(5!_0 \cdot {}^6C_4 \cdot 4!_0\right)$.

$M_1\ M_2\ M_3\ M_4$ $P_1\ P_2\ P_3$ $C_1\ C_2$

$- P_1 - P_2 - P_3 - C_1 - C_2 -$

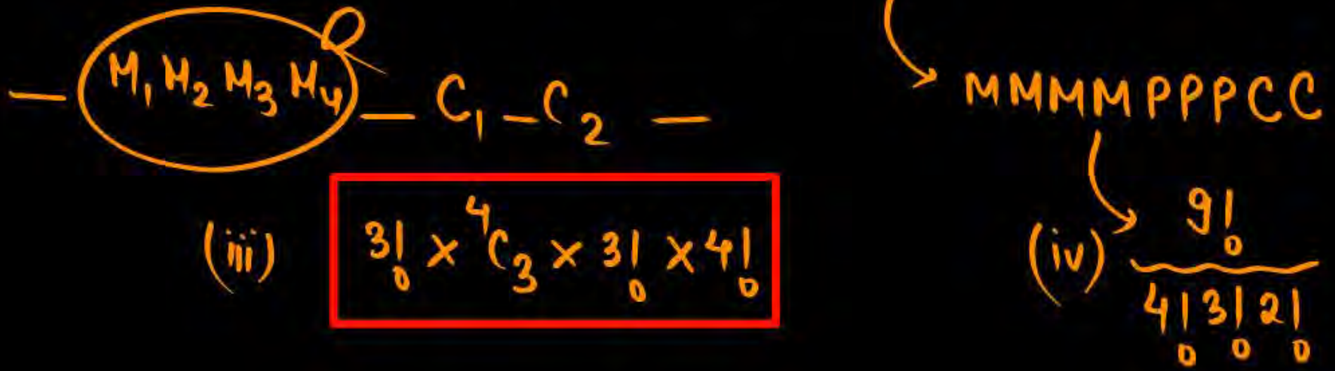
$5!_0 \times {}^6C_4 \times 4!_0$

LAST CLASS RECAP

Consider four different books of Mathematics, three different books of Physics and two different books of Chemistry. Number of ways of arranging all nine books in a row such that:

(iii) All books of Mathematics are together while no two books of Physics should be together is_____.

(iv) All possible arrangements if books of same subject are alike_____.



H.W. Discussion

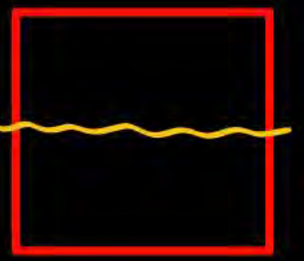
Question [JEE Mains-2023]

71%

Some couples participated in mixed double badminton tournament. If the number of matches played, so that no couple played in a match, is 840, then the total numbers of persons, who participated in the tournament, is _____.

Let n - couples participated.

Total persons = $2n$ = 16



M M F F

$$420 = {}^{n}C_2 \times {}^{n-2}C_2 \times 2$$

M_1 F_1

M_2 F_2 ✓

M_3 F_3

M_4 F_4 ✓

$$420 = \frac{n!}{2!(n-2)!} \times \frac{(n-2)!}{2!(n-4)!} = \frac{n.(n-1)(n-2)(n-3)(n-4)!}{4.(n-4)!}$$

$$\frac{n(n-1)(n-2)(n-3)}{4} = 420$$

$$n(n-1)(n-2)(n-3) = 6 \times 7 \times 5 \times (2 \times 4)$$

$$= 8 \times 7 \times 6 \times 5$$

$$n = 8$$

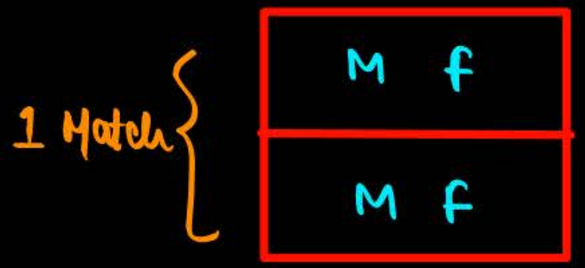
$$(n^2-3n)(n^2-3n+2) = 1680$$

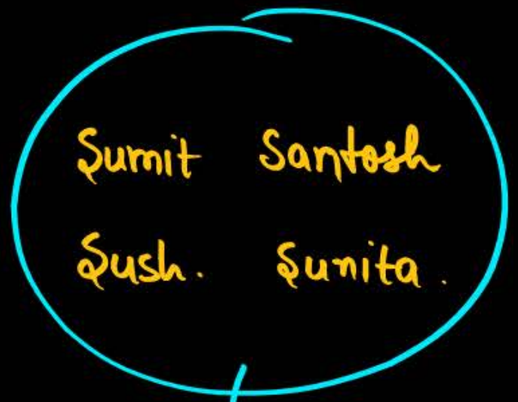
$n^2-3n = t$ ✓

$t(t+2) = 1680$

↳ $t \rightarrow$ Quad.

- Mixed double





2 Matches.

M1

Sum. & Sush.
Sant. & Sunita

M2

Sum. & Sunita.
Sant. & Sush.

Question [JEE Mains-2026]

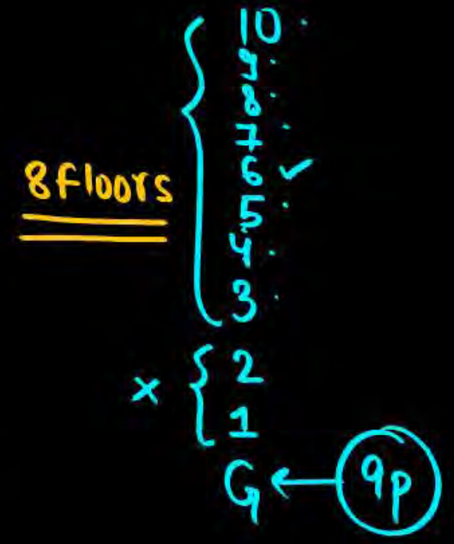
A building has ground floor and 10 more floors. Nine persons enter in a lift at the ground floor. The lift goes up to the 10th floor. The number of ways, in which any 4 persons exit at a floor and the remaining 5 persons exit at a different floor, if the does not stop at the first and the second floors, is equal to

A 2184

B 3064

C 7056

D 11340



$${}^8C_1 \times {}^9C_4 \times {}^7C_1 \times 1$$

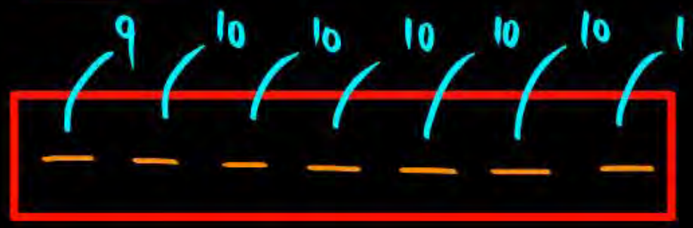
$$8 \times 126 \times 7 = 7056$$

Question [JEE Mains-2025]

[Ans. : 14]

If the number of seven-digit numbers, such that the sum of their digits is even, is $m \cdot n \cdot 10^n; m, n \in \{1, 2, 3, \ldots, 9\}$, then $m + n$ is equal to

$m + n = 9 + 5 = 14$



$\frac{9 \times 10^6}{2} = \frac{9 \times 10^5}{2}$ (5)

1000000 → 1
1000001 → 2
1000002 → 3
1000003 → 4
⋮

Total seven digit no. $= 9 \times 10^6$

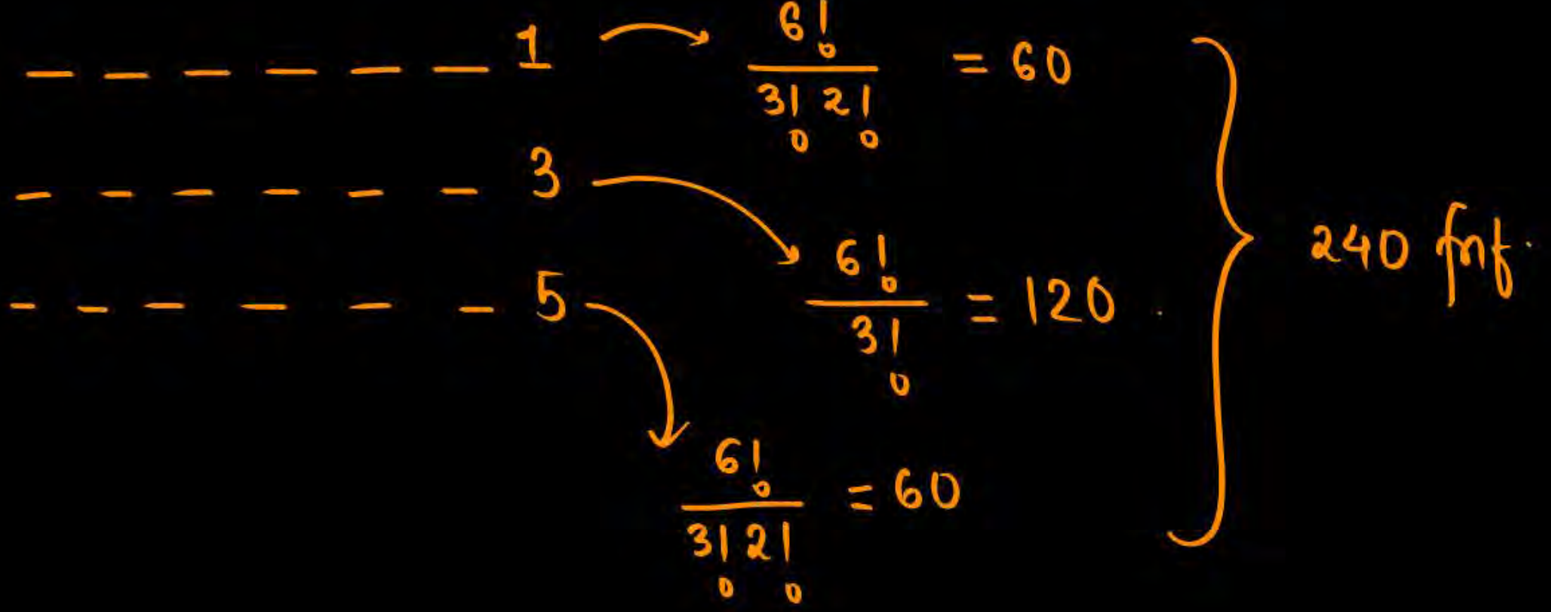
half even.

half odd

$\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

Question [JEE Mains-2023]

The number of seven digits odd numbers, that can be formed using all the seven digits 1, 2, 2, 2, 3, 3, 5 is _____.



Question [JEE Mains-2023]

The number of 9 digit numbers that can be formed using all the digits of the number 123412341 so that the even digits occupy only even places, is _____.

Speech of 7 politicians in a political rally is arranged one after other, number of ways in which they can do the same if-

(i) No Restriction

(ii) Mr. A wants to speech before Mr. B and Mr. B before Mr. C.

(iii) Mr. A wants to speech before Mr. B and Mr. B after Mr C.

A, B, C, D, E, F, G

(i) $7!$

(ii) $A > B > C$

ABC DEFG ✓

GDAEBFC

$fnf = ({}^{7}C_3 \times 4!)$

(iii) $\begin{cases} A_C_B \\ C_A_B \end{cases}$

A B C

* * *

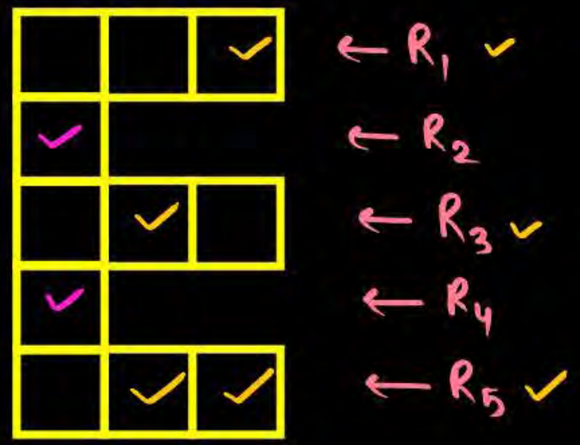
$A > B$ & $C > B$

ADCEBFG ✓

CDAEBFG ✓

$fnf = {}^{7}C_3 \times 2 \times 4!$

The symbols +, +, ×, × *, •, are placed in the squares of the adjoining figure, The number of ways of placing symbols so that no row remains empty is k, then the ten's digit of k is _____.



(Selection of squares) arrang

$$\text{Req.} = (3\times3\times3\times3)\times\left(\frac{6!}{2!\,2!\,0\,0}\right)$$

$${}^{3}C_{1}\times{}^{3}C_{2}\times{}^{3}C_{1}\times{}^{3}C_{1}$$

RANK OF A WORD IN DICTIONARY

$UNITA

Alp. order → A, ~~I~~, ~~N~$~~, T, ~~U~~.

Alphabetical Order

_ _ _ _ _ _

↳ $6!_0 = 720$

A_ _ _ _ _ → $5!_0 = 120$.

I_ _ _ _ _ → $5!_0 = 120$.

N_ _ _ _ _ → $5!_0 = 120$.

$A_ _ _ _ → $4!_0 = 24$

$I_ _ _ _ → $4!_0 = 24$

$N_ _ _ _ → $4!_0 = 24$

$T_ _ _ _ → $4!_0 = 24$

$UA_ _ _ → $3!_0$

$UI_ _ _ → $3!_0$.

$UNA_ _ → $2!_0$ ✓

$UNIAT → 1_0.

$UNITA → 1

Rank ✓

SUNITA

	S	U	N	I	T	A
	4	6	3	2	5	1
$R_1 \rightarrow$	3	4	2	1	1	0
$R_2 \rightarrow$	$5!$	$4!$	$3!$	$2!$	$1!$	$0!$

$$\text{Rank} = 3 \cdot 5! + 4 \cdot 4! + 2 \cdot 3! + 1 \cdot 2! + 1 \cdot 1! + 0 + 1$$

When All Letters are different

Find rank of word GANIT

$5! = 120$

A, G, I, N, T.

A _ _ _ _ → $4! = 24$

G A I _ _ → $2!$

G A N I T → 1

27th

When All Letters are not different

Find rank of word MASOOM

A, M, M, S, O, O.

Total = $\frac{6!}{2!2!}$

A _ _ _ _ _ _ → $\frac{5!}{2!2!} = 30$

M A M _ _ _ → $\frac{3!}{2!} = 3$

M A O _ _ _ → $3! = 6$

M A S M _ _ → $\frac{2!}{2!} = 1$

M A S O M O → 1.

M A S O O M → 1

42th.

Question

TRICK

When All Letters are different

Find rank of word GANIT

	2	1	4	3	5
	G	A	N	I	T
$R_1 \rightarrow$	1	0	1	0	0
$R_2 \rightarrow$	$4!_0$	$3!_0$	$2!_0$	$1!_0$	$0!_0$

$1 \times 4!_0 + 0 + 1 \times 2!_0 + 0 + 0 + (1)$

$24 + 2 + 1 = (27)$

When All Letters are not different

Find rank of word MASOOM

	2	1	4	3	3	2
	M	A	S	O	O	M
$R_1 \rightarrow$	1	0	3	1	1	0
$R_2 \rightarrow$	$5!_0$	$4!_0$	$3!_0$	$2!_0$	$1!_0$	$0!_0$
	$\frac{}{2!2!_0}$	$\frac{}{2!_0}$	$\frac{}{2!_0}$	$\frac{}{2!_0}$		

$= 30 + 0 + 9 + 1 + 1 + 0 + (1)$

$= 42.$

Question

Find rank of word ANURADHA

	A	N	U	R	A	D	H	A
	1	4	6	5	1	2	3	1
$R_1 \rightarrow$	0	4	5	4	0	1	1	0
$R_2 \rightarrow$	$\frac{7!_0}{3!_0}$	$\frac{6!_0}{2!_0}$	$\frac{5!_0}{2!_0}$	$\frac{4!_0}{2!_0}$	$\frac{3!_0}{2!_0}$	$2!_0$	$1!_0$	$0!_0$

$$0 + \frac{6!_0}{2!_0} \times 4 + 5\,\frac{5!_0}{2} + 4 \cdot \frac{4!_0}{2!_0} + 0 + 2 + 1 + 0 + 1$$

DIBY (Concept)

Question [JEE Mains-2020]

If the letters of the word 'MOTHER' be permuted and all the words so formed (with or without meaning) be listed as in a dictionary, then the position of the word 'MOTHER' is_____.

Trick

	3	4	6	2	1	5
	M	O	T	H	E	R
$R_1 \rightarrow$	2	2	3	1	0	0
$R_2 \rightarrow$	$5!_0$	$4!_0$	$3!_0$	$2!_0$	$1!_0$	$0!_0$

$\overset{0}{Q}IBY$

↓

by Concept

Rank = 240 + 48 + 18 + 2 + 0 + 0 + 1.

= 309

Reverse Question

If the letters of the word 'MOTHER' be permuted and all the words so formed (with or without meaning) be listed as in a dictionary, then word written at 519th position??

Concept :- E, ~~H~~, M, ~~O~~, ~~R~~, T.

E _ _ _ _ _ → $5!_0$ = 120.

H _ _ _ _ _ → $5!_0$ = 120 — 240.

M _ _ _ _ _ → $5!_0$ = 120 — 360.

O _ _ _ _ _ → $5!_0$ = 120 — 480.

RE _ _ _ _ → $4!_0$ = 24 — 504

RHE _ _ _ → $3!_0$ = 6 — 510.

RHM _ _ _ → $3!_0$ = 6 — 516

RHOE _ _ → $2!_0$ = 2 — 518.

RHOMET → 1 — 519

QIBY — ✗

(Word at 453rd ??)

Question [IIT-JEE-2007, Paper-2]

The letters of word **COCHIN** are permuted and all the permutations are arranged in an alphabetical order as in a English dictionary. The number of words that appear before the word COCHIN is

A 360

B 192

C 96

D 48



Reverse Question

If all the letters of the word PUBLIC be permuted and all the words so formed (with or without meaning) be listed as in a dictionary, then word written at 437th position??

H.W.

Question [JEE Mains-2016]

If all the words (with or without meaning) having five letters, formed using the letters of the word SMALL and arranged as in a dictionary; then the position of the word SMALL is

DIBY - (Y)

A 59th

B 52nd

C 58th

D 46th

Question [JEE Mains-2023]

If the letters of the word MATHS are permuted and all possible words so formed are arranged as in a dictionary with serial numbers, then the serial number of the word THAMS is

A 103

B 104

C 102

D 101

DIBY - (7)

Let S = {1, 2, 3, 5, 7}. The rank of 35773 if all 5 digit number formed by digits of set S, are arranged in a dictionary in ascending order and repetition of digits is allowed.

_ _ _ _ _ Total = 5^5 ✓

1 _ _ _ _ → 5^4

2 _ _ _ _ → 5^4

31 _ _ _ → 5^3

32 _ _ _ → 5^3

33 _ _ _ → 5^3

351 _ _ → 5^2

352 _ _ → 5^2

353 _ _ → 5^2

355 _ _ → 5^2

3571 _ → 5

3572 _ → 5

3573 _ → 5

3575 _ → 5

35771 → 1

35772 → 1

35773 → 1

rnf = 2×625 + 3×125 + 4×25 + 4×5 + 3.

Question Practice

Acidic Question

[Ans. : 210]

Find the number of 7 digit integers whose sum of digits is 59.

Case Making



Question

[Ans. : D]

The number of 5-digit ATM passwords having at least one of their digits repeated is



A 90,000

B 10,000

C 30,240

D 69,760

DO IT BY YOURSELF (DIBY)

DIBY-15

Eleven books consisting of 5 Mathematics, 4 Physics and 2 Chemistry are placed on a shelf. The number of possible ways of arranging them on the assumption that the books of the same subject are all together is

(A) $4!2!$ (B) $11!$ (C) $5!4!3!2!$ (D) None of these

[Ans. : (C)]

DIBY-16

The number of six letter words (with or without meaning), formed using all the letters of the word 'VOWELS', so that all the consonants never come together, is

(A) 576 (B) 140 (C) 356 (D) 209

[Ans. : (A)]

O, E, (V, W, L, S)

$Ans. = 6! - 3! \times 4!$

DIBY-17 (same as pair of shoes wala sawaal)

Six married couple are sitting in a room. Then, the number of ways in which 4 people can be selected, so that They do not form a couple, is

(A) 240 (B) 255 (C) 480 (D) 495

[Ans. : (A)]

DIBY-18

How many of the arrangements of the letter of the word "LOGARITHM" begin with a vowel and end with a consonant?

[Ans. : 90720]

DIBY-19 (Same as polit· speech wala sawaal)

Number of permutations of 1, 2, 3, 4, 5, 6, 7, 8 and 9 taken all at a time, such that the Digit 1 appearing somewhere to the left of 2
3 appearing to the left of 4 and
5 somewhere to the left of 6, is
(e.g. 815723946 would be one such permutation)

(A) $9 \cdot 7!$ (B) $8!$ (C) $5! \cdot 4!$ (D) $8! \cdot 4!$

[Ans. : (A)]

DIBY-20

A test consists of 6 multiple choice questions, each having 4 alternative answers of which only one is correct. The number of ways, in which a candidate answers all six questions such that exactly four of the answers are correct, is

[JEE Mains-2020]

[Ans. : 135]

DIBY-21

Let the number of elements in sets A and B be five and two respectively. Then the number of subsets of A × B each having at least 3 and at most 6 elements is

[JEE Mains-2023]

(A) 782 (B) 772 (C) 752 (D) 792

[Ans. : (D)]

DIBY-22

5 Indian & 5 American couples meet at a party & shake hands. If no wife shakes hands with her own husband & no Indian wife shakes hands with a male, then the number of hands shakes that takes place in the party is

(A) 95 (B) 110 (C) 135 (D) 150

[Ans. : (C)]

DIBY-23

Number of natural numbers between 100 and 1000 such that at least one of their digits is 7, is

(A) 225 (B) 243 (C) 252 (D) none

[Ans. : (C)]

DIBY-24

Let S = {1, 2, 3, ...9} for k = 1, 2, ...5. Let N_k be the number of subsets of S, each containing five elements out of which exactly k are odd.

Then $N_1 + N_2 + N_3 + N_4 + N_5$ = [JEE (Adv.)-2017]

(A) 125 (B) 210 (C) 252 (D) 126

[Ans. : (.....)]

DIBY-25

The number of seven-digit numbers, that can be formed by using the digits 1, 2, 3, 5 and 7 such that each digit is used at least once. is: [JEE Mains-2026]

(A) 15400 (B) 17800 (C) 16800 (D) 29400

[Ans. : (C)]

DIBY-26

Let P_n denote the total number of triangles formed by joining the vertices of an n-side regular polygon. If $P_{n+1} - P_n = 66$, then the sum of all distinct prime divisors of n is: [JEE Mains-2026]

(A) 7 (B) 8 (C) 5 (D) 6

[Ans. : (C)]

DIBY-27

The number of ways, of forming a queue of 4 boys and 3 girls such that all the girls are not together, is: [JEE Mains-2026]

(A) 5040 (B) 3050 (C) 3410 (D) 4320

[Ans. : (D)]

DIBY-28

A box contains 5 blue, 6 yellow and 4 red balls The number of ways, of drawing 8 balls containing at least two bails of each colour, is: [JEE Mains-2026]

(A) 4100 (B) 4140 (C) 4230 (D) 4290

[Ans. : (A)]

DIBY-29

The number of numbers greater than 5000, less than 9000 and divisible by 3, that can be formed using the digits 0, 1, 2, 5, 9, if the repetition of the digits is allowed, is [JEE Mains-2026]

[Ans. : 42]

DIBY-30

Let $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$. Let x be the number of 9-digit numbers formed using the digits of the set S such that only one digit is repeated and it is repeated exactly twice. Let y be the number of 9-digit numbers formed using the digits of the set S such that only two digits are repeated and each of these is repeated exactly twice. Then,

[JEE Mains-2026]

(A) $56x = 9y$ (B) $29x = 5y$ (C) $45x = 7y$ (D) $21x = 4y$

[Ans. : (D)]

DIBY-31

There are 12 points in a plane, no three of which are in the same straight line, except 5 points which are collinear. Then the total number of triangles that can be formed with the vertices at any three of these 12 points is

[JEE Mains-2025]

(A) 230 (B) 220 (C) 200 (D) 210

[Ans. : (D)]

DIBY-32

The number of five digit numbers, greater than 40000 and divisible by 5, which can be formed using the digits 0, 1, 3, 5, 7 and 9 without repetition, is equal to

[JEE Mains-2023]

(A) 120 (B) 132 (C) 72 (D) 96

[Ans. : (A)]

DIBY-33

The total number of three-digit numbers, divisible by 3, which can be formed using the digits 1, 3, 5, 8, if repetition of digits is allowed, is: [JEE Mains-2023]

(A) 22 (B) 18 (C) 21 (D) 20

[Ans. : (A)]

DIBY-34

If ${}^{2n+1}P_{n-1} : {}^{2n-1}P_n = 11:21$, then $n^2 + n + 15$ is equal to: [JEE Mains-2023]

[Ans. : 45]

DIBY-35

A person forgets his 4-digit ATM pin code. But he remembers that in the code all the digits are different, the greatest digit is 7 and the sum of the first two digits is equal to the sum of the last two digits. Then the maximum number of trials necessary to obtain the correct code is [JEE Mains-2023]

[Ans. : 72]

DIBY-36

The number of seven digit positive integers formed using the digits 1, 2, 3 and 4 only and sum of the digits equal to 12 is _____. [JEE Mains-2023]

[Ans. : 413]

DIBY-37

A boy needs to select five courses from 12 available courses, out of which 5 courses are language courses. If he can choose at most two language courses, then the number of ways he can choose five courses is [JEE Mains-2023]

[Ans. : 546]

DIBY-38

There are 5 points P_1, P_2, P_3, P_4, P_5 on the side AB, excluding A and B, of a triangle ABC. Similarly there are 6 points $P_6, P_7, \ldots, P_{11}$ on the side BC and 7 points $P_{12}, P_{13}, \ldots, P_{18}$ on the side CA of the triangle. The number of triangles, that can be formed using the points $P_1, P_2, \ldots, P_{18}$ as vertices, is: [JEE Mains-2024]

(A) 751 (B) 776 (C) 796 (D) 771

[Ans. : (A)]

DIBY-39

A committee of 5 is to be chosen from a group of 9 people. Number of ways in which it can be formed if two particular persons either serve together or not at all and two other particular persons refuse to serve with each other, is

(A) 41 (B) 36 (C) 47 (D) 76

[Ans. : (A)]



Homework

Re-attempt all Class Examples

DPP & DIBY

It is never about end result

It's always about Journey

Thank You

#futureIITians